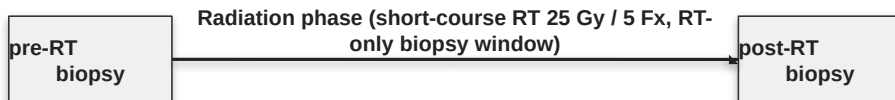


A



Static baseline predictor (Fig 5)

4-feature ElasticNet
DSB repair · frac_amp
MHC-II · MSI %

Nested AUC 0.745
95 % CI 0.56–0.90

single pre-CRT read

§3.9 Target engagement

Do the baseline predictor axes themselves move post-RT in biologically predicted direction?

RNA-paired n = 12
oriented-Δ composites

EMT good 6/6 P = 0.016
40 / 48 concordant

§3.10 IGH repertoire directional coherence

Does IGH V-gene response to RT show reproducible direction per group, even with noisy magnitudes?

RNA-paired n = 12
53 V-genes, LOO coherence

3 aggregate P < 0.05
good coherent / bad stochastic

§3.11 Cascade phenomenology

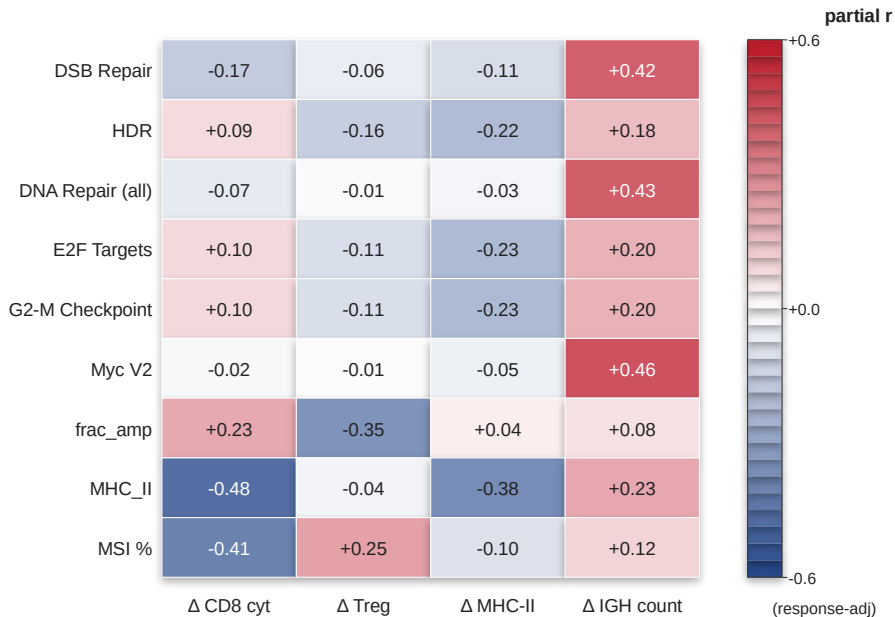
Do mutation, neoantigen, HLA-LOH, Treg, MHC-II, B-cell features show coherent pre → post Δ in good?

mixed n = 11 / 12 / 14
BCa 95 % CI, exploratory

Treg Δ only robust between-group
others within-group significant

Convergence null (§3.8)

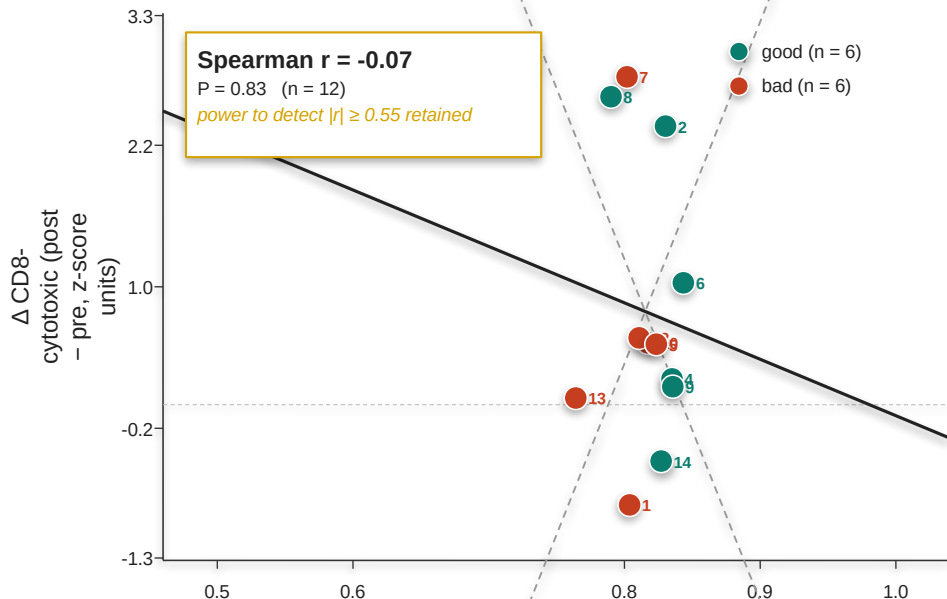
Static baseline axis ⊥ dynamic pharmacodynamic axes — 0 / 36 targeted pairs P < 0.05 (1.8 expected by chance); see Panel B and Panel C for the test and headline pair.

B**Baseline feature (pre-CRT)**

0 / 36 pairs significant at $P < 0.05$ · 0 / 36 at BH $q < 0.10$ · 1.8 expected by chance

The baseline Thread-1 predictor axes do not forecast the magnitude of the paired radiation-phase Δ .

C



Headline pair — DSB repair baseline does not forecast the post-RT CD8-cytotoxic Δ

Static baseline $\perp$ dynamic pharmacodynamic axes; the cascade in §3.11 is phenomenology, not a causal downstream.